ANALISIS DIFFERENTIALLY EXPRESSED GENES (DEG) DATASET GSE110223 BASIS IDENTIFIKASI 10 CORE GENES UNTUK DIAGNOSIS DINI CRC

Pendahuluan

Seperti dijelaskan Islam et al. (2023), DEG dari GSE110223 jadi basis identifikasi 10 core genes untuk diagnosis dini CRC. Colorectal cancer (CRC) adalah salah satu kanker paling umum dengan tingkat mortalitas tinggi. Jurnal tersebut menggunakan dataset GSE110223 (bersama GSE106582 dan GSE74602) untuk mengidentifikasi differentially expressed genes (DEGs) antara sampel kanker dan jaringan adjacent (kontrol). Analisis DEG penting untuk memahami mekanisme molekuler CRC, seperti oncogenes dan tumor-suppressor genes, yang bisa mendukung diagnosis dini dan terapi.

Untuk itu kami mencoba melakukan analisis DEG pada GSE110223 menggunakan GEO2R untuk mengidentifikasi DEGs signifikan, mengidentifikasi arah perubahan ekspresi (up-regulated dan down-regulated) dan menginterpretasikan implikasi biologisnya untuk CRC.

Metode

Dataset yang digunakan adalah GSE110223 dari NCBI-GEO. Dataset ini berisi 26 sampel: 13 sampel primary CRC (kanker kolorektal) dan 13 sampel jaringan normal adjacent (kontrol).(Islam et al., 2023)

Pembagian grup di dasarkan pada modul dan jurnal Exploring Core Genes by Comparative Transcriptomics Analysis for Early Diagnosis, Prognosis, and Therapies of Colorectal Cancer, sampel dikelompokkan menjadi dua grup di GEO2R:

1. Grup 1: Normal (13 sampel, GSM IDs: GSM2982906–GSM2982930, mewakili jaringan adjacent/normal).
2. Grup 2: Cancer (13 sampel, GSM IDs: GSM2982907– GSM2982931, mewakili jaringan kanker).

   Grouping "Normal vs Cancer" hasilkan logFC = Normal - Cancer (up-regulated in cancer: logFC < -1; down-regulated: logFC >1), sesuai konvensi jurnal (hlm. 3: |log2FC| >1 untuk signifikan).

Samples Define groups Selected 26 out of 26 samples

Columns Set

Group	Accession	Title	Source name	Tissue	Anatomic location	Presence of synchronous distant metastasis	Suv pet variable values	Vb pet variable values	K1 pet variable values	K2 pet variable values	K3 pet values
Normal	GSM2982906	Patient_1_normal_rep1_HG-U133A	Patient 1 normal adjacent sample	normal adjacent	coecum	-	0.882	0.0819	0.4168	0.7663	0.2687
Normal	GSM2982908	Patient_2_normal_rep1_HG-U133A	Patient 2 normal adjacent sample	normal adjacent	rectosigmoid	-	1.453	0.04932978	0.357459421	0.639450094	0.0455
Normal	GSM2982910	Patient_3_normal_rep1_HG-U133A	Patient 3 normal adjacent sample	normal adjacent	c.ascending	-	1.82	0.0205	0.2157	0.4506	0.0208
Normal	GSM2982912	Patient_4_normal_rep1_HG-U133A	Patient 4 normal adjacent sample	normal adjacent	c.ascending	-	1.715	0.030894	0.535926	0.921032	0.0357
Normal	GSM2982914	Patient_5_normal_rep1_HG-U133A	Patient 5 normal adjacent sample	normal adjacent	right colic flexure	-	1.444	0.018553	0.44613	0.653787	0.0126
Normal	GSM2982916	Patient_6_normal_rep1_HG-U133A	Patient 6 normal adjacent sample	normal adjacent	rectum	-	2.602	0.0495	0.0919	0.0833	0.0015
Normal	GSM2982918	Patient_7_normal_rep1_HG-U133A	Patient 7 normal adjacent sample	normal adjacent	sigmoid	-	1.278	0.160428	0.361181	0.513034	0.0304
Normal	GSM2982920	Patient_8_normal_rep1_HG-U133A	Patient 8 normal adjacent sample	normal adjacent	right colic flexure	-	1.136	0.085693	0.502894	0.785571	0.0102
Normal	GSM2982922	Patient_9_normal_rep1_HG-U133A	Patient 9 normal adjacent sample	normal adjacent	coecum	-	1.405	0.022322	0.333216	0.597589	0.0696
Normal	GSM2982924	Patient_10_normal_rep1_HG-U133A	Patient 10 normal adjacent sample	normal adjacent	right colic flexure	-	2.25	0.0132	0.3612	0.615	0.0085
Normal	GSM2982926	Patient_11_normal_rep1_HG-U133A	Patient 11 normal adjacent sample	normal adjacent	rectum	-	1.612	0.009284	0.376493	0.984928	0.1164
Normal	GSM2982928	Patient_12_normal_rep1_HG-U133A	Patient 12 normal adjacent sample	normal adjacent	left colic flexure	-	1.757	0.0385	0.3427	0.4974	0.0354

Samples Define groups Selected 26 out of 26 samples

Columns Set

Normal	GSM2982926	Patient_11_normal_rep1_HG-U133A	Patient 11 normal adjacent sample	normal adjacent	rectum	-	1.612	0.009284	0.376493	0.984928	0.1164
Normal	GSM2982928	Patient_12_normal_rep1_HG-U133A	Patient 12 normal adjacent sample	normal adjacent	left colic flexure	-	1.757	0.0385	0.3427	0.4974	0.0354
Normal	GSM2982930	Patient_13_normal_rep1_HG-U133A	Patient 13 normal adjacent sample	normal adjacent	sigmoid	-	1.951	0.028	0.1112	0.0877	0.0625
Cancer	GSM2982907	Patient_1_cancer_rep1_HG-U133A	Patient 1 primary colorectal adenocarcinoma	primary colorectal adenocarcinoma	coecum	No	19.4544	0.075227	0.380527	0.868753	0.2151
Cancer	GSM2982909	Patient_2_cancer_rep1_HG-U133A	Patient 2 primary colorectal adenocarcinoma	primary colorectal adenocarcinoma	rectosigmoid	No	13.019	0.045227844	0.354902684	0.73648247	0.1814
Cancer	GSM2982911	Patient_3_cancer_rep1_HG-U133A	Patient 3 primary colorectal adenocarcinoma	primary colorectal adenocarcinoma	c.ascending	No	4.25	0.070135	0.280101	0.519583	0.0758
Cancer	GSM2982913	Patient_4_cancer_rep1_HG-U133A	Patient 4 primary colorectal adenocarcinoma	primary colorectal adenocarcinoma	c.ascending	No	9.237	0.036756	0.467087	0.664059	0.1519
Cancer	GSM2982915	Patient_5_cancer_rep1_HG-U133A	Patient 5 primary colorectal adenocarcinoma	primary colorectal adenocarcinoma	right colic flexure	No	1.759	0.025359	0.364152	0.999	0.0436
Cancer	GSM2982917	Patient_6_cancer_rep1_HG-U133A	Patient 6 primary colorectal adenocarcinoma	primary colorectal adenocarcinoma	rectum	Yes	6.52	0.100303	0.83771	0.958854	0.0642
Cancer	GSM2982919	Patient_7_cancer_rep1_HG-U133A	Patient 7primary colorectal adenocarcinoma	primary colorectal adenocarcinoma	sigmoid	No	4.968	0.182375	0.341884	0.501936	0.0956
Cancer	GSM2982921	Patient_8_cancer_rep1_HG-U133A	Patient 8 primary colorectal adenocarcinoma	primary colorectal adenocarcinoma	right colic flexure	No	2.37	0.124744	0.393569	0.678676	0.0572
Cancer	GSM2982923	Patient_9_cancer_rep1_HG-U133A	Patient 9 primary colorectal adenocarcinoma	primary colorectal adenocarcinoma	coecum	Yes	12.4	0.084263	0.564397	0.784526	0.1626
Cancer	GSM2982925	Patient_10_cancer_rep1_HG-U133A	Patient 10 primary colorectal adenocarcinoma	primary colorectal adenocarcinoma	right colic flexure	Yes	3.03	0.017039	0.248712	0.989947	0.0965
Cancer	GSM2982927	Patient_11_cancer_rep1_HG-U133A	Patient 11 primary colorectal adenocarcinoma	primary colorectal adenocarcinoma	rectum	No	5.686	0.06063329	0.390804794	0.86060478	0.1082
Cancer	GSM2982929	Patient_12_cancer_rep1_HG-U133A	Patient 12 primary colorectal adenocarcinoma	primary colorectal adenocarcinoma	left colic flexure	No	2.488	0.014924	0.330671	0.759424	0.0229
Cancer	GSM2982931	Patient_13_cancer_rep1_HG-U133A	Patient 13 primary colorectal adenocarcinoma	primary colorectal adenocarcinoma	sigmoid	Yes	10.24	0.00426	0.876192	0.852647	0.1149

Parameter utama GEO2R: Metode statistik LIMMA (Linear Models for Microarray Data) dengan koreksi Benjamini-Hochberg untuk adjusted p-value (adj.P.Val). Cut-off: adj.P.Val < 0.05 dan |log2FC| >1 (upregulated jika log2FC < -1 untuk cancer; downregulated >1). Analisis default: Auto-detect annotation platform, apply limma precision weights (vooma), tanpa normalisasi ekstra karena data pre-processed (modul hlm. 3: mirip GPL570 di GSE18090).

$$DEGs_g = \begin{cases} \text{DEG (Upregulated)}, & \text{if } \ \text{adj.}p\text{.value} < 0.05 \text{ and } \text{Log}_2\text{FC}_g > +1.0 \\ \text{DEG (Downregulated)}, & \text{if } \ \text{adj.}p\text{.value} < 0.05 \text{ and } \text{Log}_2\text{FC}_g < -1.0 \end{cases}$$

Skema replikasi: (1) Akses GEO via modul hlm. 1 (search "GSE110223" di "Keyword or GEO Accession"); (2) Buka GEO2R, define groups seperti atas; (3) Jalankan "Analyze" (LIMMA default); (4) Download full table TSV (~22k probes); (5) Filter DEG via Excel, (6) mengurutkan hasil berdasarkan P-Value terkecil; (7) Identifikasi DEG signifikan (adj.P.Val 1 = fold change ≥2x) via Excel/Python; (8) Hindari batch effect via paired design GSE110223. (Islam et al., 2023)

GEO2R | Options | Profile graph | R script

Apply adjustment to the P-values. More...

- (●) Benjamini & Hochberg (False discovery rate)
- ( ) Benjamini & Yekutieli
- ( ) Bonferroni
- ( ) Holm

Apply log transformation to the data. More...

- (●) Auto-detect
- ( ) Yes
- ( ) No

Apply limma precision weights (vooma). More...

- (●) Yes
- ( ) No

Force normalization. More...

- ( ) Yes
- (●) No

Category of Platform annotation to display on results.

- ( ) Submitter supplied
- (●) NCBI generated

Plot displays. More...

Significance level cut-off (enter number between 0 and 1) [0.05]

Log 2 fold change threshold [0]

Volcano and Mean-difference plot contrasts (select up to 5)
0 selected (clear)

- [ ] Normal vs Cancer

If you edit *Options* after performing an analysis, click *Reanalyze* to apply the edits:

Reanalyze

Hasil

GEO2R hasilkan ~22k probes; filter dapat 736 DEG (448 down in cancer/logFC>1; 288 up/logFC<-1).

Arah Ekspresi	ID	logFC	Gene.symbol	Gene.title
Upregulation	206784_at	4,50001139	AQP8	aquaporin 8
	206209_s_at	4,38429410	CA4	carbonic anhydrase 4
	220834_at	4,31390910	MS4A12	membrane spanning 4-domains A12
	220026_at	4,12318681	CLCA4	chloride channel accessory 4
	214142_at	3,93662118	ZG16	zymogen granule protein 16
	205950_s_at	3,86076895	CA1	carbonic anhydrase 1
	205097_at	3,83814265	SLC26A2	solute carrier family 26 member 2
	203908_at	3,63506784	SLC4A4	solute carrier family 4 member 4
	207003_at	3,48269732	GUCA2A	guanylate cyclase activator 2A
	217546_at	3,42552394	MT1M	metallothionein 1M
Downregulation	204475_at	-3,11760357	MMP1	matrix metallopeptidase 1
	205828_at	-2,40111333	MMP3	matrix metallopeptidase 3
	203962_s_at	-2,26577192	NEBL	nebulette
	212942_s_at	-2,21300836	CEMIP	cell migration inducing hyaluronan binding protein
	204855_at	-2,20285001	SERPINB5	serpin family B member 5
	201506_at	-2,13638450	TGFBI	transforming growth factor beta induced
	202859_x_at	-2,13058354	CXCL8	C-X-C motif chemokine ligand 8
	209875_s_at	-2,09092940	SPP1	secreted phosphoprotein 1
	204470_at	-2,07186204	CXCL1	C-X-C motif chemokine ligand 1
	203256_at	-2,02492791	CDH3	cadherin 3

Hasil DEG GSE110223 kait erat dengan kondisi biologis CRC: Down-regulated genes (logFC >1, e.g., AQP8 +4.5, ADH1B +2.27) tunjuk loss tumor-suppressor di jaringan tumor vs adjacent normal—ganggu homeostasis usus (AQP8: osmoregulasi mukosa hilang, tingkatkan permeabilitas & inflamasi kronis); ADH1B: kurang detoks alkohol, promosi mutasi DNA (respon sel: apoptosis terganggu, selaras penelitian Islam (2023) pada jurnal Exploring Core Genes by Comparative Transcriptomics Analysis hlm. 2 suppressor loss). Up-regulated (logFC <-1, e.g., MMP1 -3.12, CDH3 -2.02) jadi oncogenes: MMP1 degradasi matriks ekstraseluler (ECM), dukung invasi/metastasis (respon sel: EMT aktif); CDH3 ubah adhesi, bantu migrasi tumor (pathway Wnt/cell cycle enriched, mirip core genes TOP2A di TSV full). Secara keseluruhan, ini implikasikan perlakuan awal CRC (inflamasi → proliferasi), potensi biomarker (test expr ADH1B untuk stadium dini) & terapi (inhibitor MMP1 seperti marimastat.(Islam et al., 2023)

Kesimpulan

Analisis identifikasi 736 DEG, replikasi jurnal dengan down DEG dominan (e.g., AQP8 loss suppressor) untuk diagnosis dini CRC. Implikasi: Target terapi (inhibitor MMP1); prognosis via stage expr. Limitasi: Single dataset—lanjut integrasi GSE106582 (jurnal hlm. 2). Saran: Enrichment full via STRING untuk core genes.

Daftar Pustaka

Edgar R, Domrachev M, Lash AE. Gene Expression Omnibus: NCBI gene expression and hybridization array data repository
Nucleic Acids Res. 2002 Jan 1;30(1):207-10

Islam, M. A., Hossen, M. B., Horaira, M. A., Hossen, M. A., Kibria, M. K., Reza, M. S., Tuly, K. F., Faruqe, M. O., Kabir, F., Mahumud, R. A., & Mollah, M. N. H. (2023). Exploring Core Genes by Comparative Transcriptomics Analysis for Early Diagnosis, Prognosis, and Therapies of Colorectal Cancer. *Cancers*, *15*(5). https://doi.org/10.3390/cancers15051369

OmicsLite. (2025). Modul Week 2: Eksplorasi Data & Analisis Praktis dengan GEO2R.